

# The rest, if you ask.

Answers, definitions, and **where to point**.

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## How to answer

One question, about ninety seconds. The judges' criterion in their own words: how effectively the technology can support scientists, improve lab workflows, or advance biological discovery. Every answer should land on one of those three. Every answer has the same shape: agree with the true part, give the number, end on what a lab does on Monday.

- **Agree first.** Most hard questions contain a true observation. Say "yes, and" before "but"; it costs nothing and disarms the room.
- **One number, said once.** Each answer below has a single figure to reach for. Don't stack three.
- **End on the lab.** The criterion is usability. Finish with the action a lab takes, not with the theory.
- **Never say** *federated, corpus, moat, or proprietary data*. Say *shared reference, every lab running that line, the field*.
- **Never claim** the quarter is proven (one institution so far), that brightfield works (not tested), or that instrument and cells are separated (not yet).
- If you don't know, say which experiment answers it. That is a better answer than a guess from a scientist.

## Where to point

Every likely question has a board. Say the one-line answer, then open the board; A0 is the index on screen if you lose track.

| IF THEY ASK                         | ONE LINE                                                                                                        | BOARD |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------|
| Why not trust the plate's controls? | The plate's state cancels from the number, never from the cells.                                                | A1    |
| Show me the math.                   | Six operations on one vector, each a mean difference or a held-out classifier.                                  | A2    |
| Why not batch-correct it away?      | Centring a plate on its own controls takes 99.9% to 24%: it removes most of the coordinate a certificate needs. | A3    |
| Is the model the moat?              | No. Counting cells matches models on many benchmarks; the reference is the moat.                                | A4    |
| Why not just count cells?           | Same count, different texture: counts move only when cells die or divide.                                       | A5    |
| How often do I re-certify?          | Every quarter at most. In one lab the baseline is stale within weeks and still moving after thirteen.           | A6    |
| Isn't this STR?                     | STR says who; images say how. Complement, not rival.                                                            | A7    |
| Is the assay itself working?        | Positive controls certify the assay; JUMP plates carry eight.                                                   | A8    |
| Does it change the result?          | That's e8, pending; if not, the pitch narrows to QC.                                                            | A9    |
| Does it work without stains?        | Shown in fluorescence; brightfield is the bet, and the test is named.                                           | A10   |
| Why would a lab upload?             | Useful on day one against JUMP; better with every lab.                                                          | A11   |
| What does the money build?          | A reference built to break, graded against planted failures.                                                    | A12   |

# New from the 25 Sept practice

Questions the practice judge raised or set up. Each answer agrees with the true part first, gives one number, and ends on what a lab does.

## So this is a nine or ten billion dollar problem?

Not quite, and I'd rather be precise. About \$28B a year of US preclinical research isn't reproducible, and about 36% of that traces to reagents and reference materials, which includes antibodies, not only cells [12]. Cells are the input in that slice that nobody certifies against anyone else.

## Why now?

Two things changed. A public cross-lab reference exists: JUMP, eleven sites, about 116,750 compounds, free under CC0 [3]. And embeddings became a commodity: counting cells alone matches rich profiles on many bioactivity benchmarks [2]. So the scarce thing isn't the model; it's the comparison, and that's what we hold. *Board A4.*

## If someone gave you a blank check, what would you build?

A reference designed to break: many lines, plate types and objectives, with focus failures, bubbles, misaligned wells and bad metadata planted on purpose. Every normalization method gets graded on whether it keeps the lab coordinate and catches each failure. The proof of concept on public data tells us which variables to spend on. *Board A12.*

## Isn't this your former employer's IP?

*Answer only with the one sentence you have agreed with counsel. Do not improvise, and do not expand on how anything was solved at Axiom. [YOUR AGREED SENTENCE]*

## What does it cost?

Per line, per quarter. The figure isn't set; the proof of concept sets it. The quarter is a starting cadence: in the one lab measured, a baseline is stale within weeks. *Board A11.*

## Are outliers just noise to you?

No, they're the product. At Axiom, when we clustered outliers they named themselves: air bubbles, focus failures, wells clipped by misalignment. Every one is a flag a lab would want before it screens.

## Could this go beyond QC?

Yes, later: training data for microscope makers, gaps in the cell-line catalogue, and early warning when a line starts to drift. I'd frame all three as drift detection, not new biology, until the data says otherwise.

## Who's your first customer?

Any institution already holding U2OS control images, because the public reference is U2OS. *Don't name a lab until it has agreed.*

## Key definitions

Plain words first, precise words second. Use the plain ones on stage.

| TERM                             | PLAIN                                                           | PRECISE                                                                                                                                                                             |
|----------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Control well, vehicle well       | Cells with the solvent but no drug. The plate's zero.           | DMSO-treated negative control; in JUMP, columns 2 and 23 of a 384-well compound plate, 32 wells.                                                                                    |
| DMSO                             | The solvent drugs are dissolved in.                             | Dimethyl sulfoxide; hygroscopic, so its water content changes with handling.                                                                                                        |
| Positive control                 | A drug with a known effect, run on every plate.                 | JUMP uses eight compounds in columns 1 and 24; FCCP is the proposed mitochondrial one.                                                                                              |
| Cell Painting                    | Six dyes, five colour channels, one photo per part of the cell. | Multiplexed fluorescent stains for DNA, RNA, ER, actin/Golgi/membrane and mitochondria; about 1,500 features per cell [6].                                                          |
| Profile, features                | The list of numbers describing one well.                        | 3,773 measurements per well here, aggregated from single cells.                                                                                                                     |
| Batch effect                     | The lab, the day and the machine leaking into the numbers.      | Systematic non-biological variation across sites, batches, plates and wells.                                                                                                        |
| Batch correction, whitening, TVN | Scrubbing the lab out of the data.                              | Typical Variation Normalization: PCA-whitening fit on control wells so that they become spherical. Not run here; centring each plate on its own controls takes lab identity to 24%. |
| Balanced accuracy                | Average hit rate across the eleven labs.                        | Mean per-class recall; chance is $1/K = 9.1\%$ for $K = 11$ .                                                                                                                       |

| TERM                    | PLAIN                                                                       | PRECISE                                                                                                                              |
|-------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Held out by plate       | The test plates were never seen in training.                                | Grouped cross-validation on plate ID; blocks memorising a plate.                                                                     |
| Permutation test        | Shuffle the labels; a real signal collapses to chance.                      | Permuted-label balanced accuracy 9.3% against 9.1% chance.                                                                           |
| Conditional expectation | What normal looks like given the conditions.                                | $E[x \mid \text{line, lot, passage, operator, site, day}]$ ; the object the certificate estimates.                                   |
| Drift                   | Cells slowly stop behaving like they did.                                   | Distance between plate centroids grows with the time between batches and is still rising after thirteen weeks (one lab, 219 plates). |
| Passage                 | How many times the culture has been split.                                  | A covariate; ATCC advises re-authenticating after 10 passages.                                                                       |
| STR profiling           | A DNA fingerprint proving which line you have.                              | Short tandem repeat typing; identity, not behaviour.                                                                                 |
| The field               | Every lab running the same line under the same conditions.                  | The reference distribution the certificate compares against.                                                                         |
| Certificate             | A pass/fail that your cells still sit with the field, good for one quarter. | Distance from your control distribution to the field's, conditioned on covariates, in units of the field's own spread.               |
| Known-answer test       | Give the assay a question you know the answer to.                           | Positive-control effect vector agrees with the field's after conditioning; certifies the assay, not the cells.                       |

## Science and validity

Isn't this just batch effect? Everyone knows about that.

Yes, and everyone removes it. The published JUMP benchmark tested ten correction methods and found Harmony best at removing it [5]. I'm saying the thing being removed is a measurement of your lab's state. On our data, centring each plate on its own controls drops lab identity from 99.9% to 24% (white paper Fig. 1; board A3). On Monday, that's the coordinate a lab can no longer see.

## **Couldn't it be the microscope, not the cells?**

Partly. The signature is the whole pipeline: instrument, staining, handling and cells. It is not the microscope model alone: the four labs on a CV8000 are told apart as well as all eleven, at 99.99%. Image-quality features alone also name the lab, so I say "the lab", not "the cells". What matters to a customer is that position predicts the answer: inside every lab, plates whose controls sit further from the field disagree more about the same compound (Spearman 0.35, shuffled 0.01; results/e8-redteam).

## **Why that lab? Is it representative?**

It's the clearest case, not the average. Lab 2 has 13 batches across four months, so its drift can be followed. Its October batch has a normal cell count, 182 per well, so no local rule fires, and sits 7.0 spreads from the lab's own June baseline against 4.5 from lab 8. Every one of its wells is still named lab 2: identity intact, position moved. Across the field, 44 of 1,872 plates sit nearer another lab's centre than their own. An earlier version of this deck said one plate's 384 wells classified as another lab. That did not reproduce and is withdrawn; all 384 are named lab 2 (analysis/README).

## **This is fluorescence. You said label-free.**

Correct. The result is on Cell Painting, the public data that exists. Label-free brightfield is the deployment bet because it costs a lab nothing extra, and JUMP's pilot plates carry brightfield planes next to the fluorescence, so the test is one run away [3] (board A10).

## **Surely a good lab catches this with its own QC?**

Only inside the box it can see. A control well catches a bad well. Plate-level QC catches a bad plate, as long as the rest of the plate is fine. Batch-level QC catches a bad batch against that lab's own history. Nothing inside a building catches the building: if the whole site has drifted, every local check passes, because every local reference drifted with it. That is what slide 05 shows on one real lab: cell counts in range from June to October, while its October plates sit further from its own June baseline than from another lab (white paper §3). ML calls this domain shift and made it a benchmark with RxRx1 [4]. The reference has to sit outside every box.

## **Why would control wells drift in a way that matters to a drug result?**

Because every result is treated minus control on the same plate. If the state of the plate shifts both, the difference reads out the drug effect conditional on a state nobody measured. Morphology moves before viability does; our hepatocyte paper saw image profiles catch bioactivity at lower doses than the toxicity

assays, across 1,085 compounds at 8 doses [1]. Drift shows up first in exactly the place the subtraction erases.

## **Cells vary. Isn't some of this real biology, not error?**

Yes, and that's the product. Identity testing tells you the line is what it says; ATCC's own guidance cites LNCaP cells at low and high passage responding differently to the same hormones [9]. Same fingerprint, different cells. The certificate doesn't call variation an error; it tells you where you sit relative to everyone else running that line.

## **Machine learning**

### **Is the 99.9% leakage?**

Held out by plate, so no test plate was seen in training; holding out whole batches gives 99.6%. Shuffle the labels and it scores 9.3% against 9.1% chance, so the chance line is real. Drop the two labs that use 1536-well plates and the remaining nine score 99.98%, so it isn't plate format. Remove position and count features and nothing changes (results/e1-redteam).

### **What's the metric, exactly?**

Balanced accuracy: mean recall across the eleven labs, so chance is one over eleven. Raw z-scored features reach 99.9%; centring each plate on its own controls drops that to 24%, and scaling as well to 19% (white paper Fig. 1). For scale: the JUMP batch-correction benchmark scored ten such methods on this dataset [5].

### **Isn't this a solved problem in ML? Domain shift, CORAL, invariant learning.**

It's a known problem, which is the point: RxRx1 is a WILDS benchmark, and the original Kaggle competition handed competitors the control-well labels precisely so they could align domains [4]. The ML world treats the lab as a nuisance to be made invariant to. For discovery that's right. For QC it's backwards: you want to *measure* the shift, not erase it.

### **Why not just use a foundation model?**

We will; the extractor barely matters. In our hepatocyte paper, CellProfiler, a Cell Painting CNN and a pretrained vision transformer performed similarly [1]. Recursion's OpenPhenom model card itself says to batch-correct its embeddings with TVN fit on control images before they're useful [7], which is the operation that deletes the coordinate. The model is a commodity; the conditioned reference is the asset.

### **How do you know the model isn't learning something trivial?**

Sometimes it is, and I've published on exactly that: on standard bioactivity benchmarks, counting cells matches rich profiles far more often than people assume [2]. That's why the certificate is a distance in the field's own spread, not a classifier score.

## How do you actually measure a morphology change?

Two ways, and I've used both. Distance-based: the distance of a well's embedding from the vehicle mean, with an exponential saturation normalisation so one enormous move doesn't swamp the score. Classifier-based: a support-vector classifier separating treated from vehicle, which works far better once embeddings are aligned across batches with TVN or CORAL; the JUMP benchmark compared ten alignment methods of this kind [5]. Both are right for discovery, where you want invariance to the lab. Neither is right for QC, because the alignment step deletes the lab coordinate the certificate needs. Same vector, two questions; the difference is whether you align first.

## What's the classifier, exactly?

Say it as a method card, in this order, and say it plainly: it is the least interesting part and the most checkable.

| ITEM      | WHAT TO SAY                                                                                  | STATUS                      |
|-----------|----------------------------------------------------------------------------------------------|-----------------------------|
| Data      | 93,228 untreated wells, 1,872 plates, 11 labs, JUMP cpg0016, CC0                             | in the deck                 |
| Features  | 3,773 CellProfiler well-level features, z-scored on the training folds                       | in the deck                 |
| Model     | <i>your model type and regularisation, e.g. logistic regression with L2, or a linear SVM</i> | <b>fill from results/e1</b> |
| Split     | held out by plate: no test plate seen in training; <i>number of folds</i>                    | folds: <b>fill</b>          |
| Metric    | balanced accuracy = mean recall over 11 labs; chance 1/11 = 9.1%                             | in the deck                 |
| Result    | 99.9%                                                                                        | in the deck                 |
| Nulls     | permuted labels 9.3%; without the 1536-well labs 99.98%; whole batches held out 99.6%        | in the deck                 |
| Ablations | plate-centred 24%; plate-centred and scaled 19% (29,943 wells)                               | in the deck                 |

| ITEM         | WHAT TO SAY                                                                               | STATUS |
|--------------|-------------------------------------------------------------------------------------------|--------|
| Not done yet | instrument separated from site; brightfield; more than one institution for the shelf life | say so |

If asked why not a deep model: because a linear model on named features is the claim that survives cross-examination; the extractor barely matters on the hepatocyte data, and a judge can rerun a linear model that night. If the model is not linear, say what it is and why.

## Usability in a real lab

### What does a lab actually do on Monday?

Export the control-well images it already has and send them. No new assay, no dye, no instrument, no compounds and no results leave the building. The answer comes back as one of two words and a distance.

### Why would anyone share data?

They share untreated control wells only: cells plus solvent. There's no compound, no target, no result in them. And the answer each lab gets back improves with every lab that joins, so the incentive runs the right way.

### How often?

At least every quarter. In one lab's data, the distance between two plates grows from 0.63 in the same batch to 3.09 at six to thirteen weeks and 4.7 beyond, and had not levelled off in four months (white paper Fig. 2; slide 05, click 3). An earlier version said it plateaus; on the full data it does not. It's one lab and 219 plates, so it gets re-measured at every partner.

### How is this different from STR testing?

STR proves who your cells are. This proves how they're doing. ATCC tells labs to re-test identity on receipt, after ten passages and when in doubt [9]; the certificate slots into the same habit, with a behavioural line under the identity line.

### Does it work for my cell line?

It works for any line more than one lab runs. The first certificates are for U2OS, because JUMP already provides the field for it. Each new line needs a few labs to seed the reference; that's the ask.

### What about primary cells or iPSC-derived lines?

Same logic, harder conditioning: donor becomes a covariate. It's later, not never. I've built assays in primary and hiPSC-derived neurons [8], so I know what that conditioning costs.

### **Can I run this on my Incucyte / label-free system?**

That's the bet. Sartorius's own label-free classifier is trained on the plate's own control wells [10], which is the loop slide 03 describes. The images those instruments already capture are the cheapest possible input.

### **What happens when I fail?**

You get "has drifted" and a distance, and the positive-control test tells you whether it's the cells or the assay. Vehicle fails: fix the cells. Positive control fails: fix the assay. Both pass and you're done.

## **Business, market and competition**

### **Who pays, and how much?**

*Your call; the script assumes a subscription per cell line per quarter.* First customers are labs and CROs whose results have to transfer: multi-site programs, tech transfer, CRO-to-sponsor handoffs. Those are the people who already lose weeks to "it worked in our hands".

### **Why isn't this a feature Sartorius or Recursion adds next year?**

They can copy the analytics; they can't hold the reference. The holder of a reference has to be neutral, and both sell the thing the certificate might blame: the instrument, or a drug program built on the data. Recursion's public model card tells users to correct the lab out with TVN [7]; Sartorius normalises to your own wells [10]. Neither is positioned to certify.

### **Why isn't JUMP or the Broad the natural owner?**

They built the evidence, and I'd build on them, not against them. A consortium publishes an atlas; it doesn't run a per-lab, per-quarter service or stand behind an answer for a specific lab.

### **Why isn't this ATCC?**

ATCC distributes the cells it authenticates. That's fine for identity and a conflict for behaviour. They're the natural partner for a combined identity-plus-behaviour certificate, and their STR service already runs on mailed FTA cards with a recurring cadence [9].

### **How big is this?**

Irreproducible preclinical research in the US alone was estimated at about US\$28B a year, with reagents and reference materials, including cell lines, the largest category [12]. I'm not claiming that number; I'm

claiming the cell-line slice of it is the part nobody certifies.

### **What's the network effect, concretely?**

Each lab that joins tightens the field's estimate of normal for that line, which makes every other lab's certificate more precise, which makes journals, CROs and sponsors more likely to ask for it. The loop only turns if nobody fears the holder.

## **Team, risk and curveballs**

### **Why you?**

I've built imaging assays at AbbVie, insitro and Axiom, I'm second author on the hepatocyte Cell Painting paper [1], and I've coauthored the work on how this field checks its own benchmarks [2]. This is the thing I kept needing and nobody sold.

### **Why now?**

JUMP made cross-lab control data public [3], and feature extraction has become a commodity [1][7]. The reference is the only missing piece, and it's the one nobody neutral is building.

### **What's the biggest risk?**

Cold start. A reference without labs is nothing. That's why the first certificates are for U2OS, where JUMP already provides the field, and why the ask tonight is partner labs rather than money.

### **What if the signal is mostly instrument?**

Then the certificate still tells a lab its images don't sit with the field, which is still the thing a collaborator needs to know before trusting a result. And the nested run separates instrument from lab for the models JUMP shares across sites [11].

### **Have you talked to labs?**

*Answer honestly. If you have a partner, name them. If not: "Three conversations in progress; the first named partner is what I'm here for."*

### **Isn't "nobody has ever checked" an overclaim?**

Fair. Batch effects have been checked to death as noise [5]. What nobody has done is certify a lab's controls against the field's, and certify is the word on the slide.

### **What would make you wrong?**

If the same plate run in two buildings on two instruments still classifies by building after conditioning on instrument, the cells carry the signal and the product stands. If it classifies by instrument alone, the product becomes an instrument-QC certificate, which is smaller but still real. Bench time in a second building is the experiment, and it's on the ask.

## References

Every bracketed number above points here. All were checked against the source this session; unbracketed numbers come from your results files (e1, e1b, e2).

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